AI-POWERED CYBERBULLYING DETECTION AND PREVENTION SYSTEM IN SOCIAL MEDIA USING PYTHON

(CyberShield AI)

A Project Report submitted in partial fulfillment of the requirements for the award of the degree of

BACHELOR OF TECHNOLOGY
in
COMPUTER SCIENCE AND ENGINEERING

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ACADEMIC YEAR: 2025 – 2026

BONAFIDE CERTIFICATE

This is to certify that this project report entitled '**AI-POWERED CYBERBULLYING DETECTION AND PREVENTION SYSTEM IN SOCIAL MEDIA USING PYTHON**' is the bonafide work carried out by **[STUDENT NAME]** (Reg. No: **[REGISTER NUMBER]**), who carried out the project work under my supervision in partial fulfillment of the requirements for the award of the degree of Bachelor of Technology in Computer Science and Engineering during the academic year 2025-2026.

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INTERNAL EXAMINER	**EXTERNAL EXAMINER**

ABSTRACT

The exponential growth of modern social media networks has catalyzed unprecedented interpersonal communication while concurrently accelerating digital hostility, cyberbullying, hate speech, and violent harassment. Traditional content moderation systems depend either on rigid, easily bypassed static keyword filters or reactive human annotator queues that suffer from severe fatigue and high cognitive latency. This academic project presents **CyberShield AI**, an intelligent, full-stack, automated cyberbullying detection and prevention system implemented using Python 3.14, Django 5.2, Natural Language Processing (NLP), and Scikit-Learn.

The architecture implements a specialized linguistic preprocessing pipeline tailored for unstructured digital content, incorporating emoji-to-semantic token mapping, character elongation normalization (e.g., 'looooser' to 'loser'), social mention/URL filtering, and negation-preserving lemmatization. Feature extraction is driven by Sublinear Term Frequency-Inverse Document Frequency (TF-IDF) unigrams and bigrams. The system implements a 7-class taxonomy comprising *Normal/Safe*, *Harassment*, *Hate Speech*, *Threatening Language*, *Personal Attack*, *Sexual Harassment*, and *Toxic/Abusive Language*. Multiple supervised machine learning algorithms—Logistic Regression, Calibrated Linear Support Vector Machine (LinearSVC), Multinomial Naive Bayes, and Random Forest—were rigorously trained and evaluated under stratified 80/20 train/test protocols. Empirical evaluation confirmed that Logistic Regression with class-weighted balancing achieved the superior weighted F1-score of 0.6309 and accuracy of 62.07% across held-out test splits.

To ensure transparency, an Explainable AI (XAI) subsystem extracts triggering lexical tokens, computes multi-class probability scores, and provides severity-graded risk cards (Green, Orange, Red) with actionable safety protocols. The solution is delivered as an enterprise-grade Django web application featuring real-time asynchronous REST APIs (/api/analyze/), multi-line batch and CSV dataset processing, Chart.js operational dashboards, and an incident reporting center. CyberShield AI bridges academic machine learning research with mission-critical digital safety operations, empowering educational institutions, gaming servers, and online communities to foster humane digital spaces.

CHAPTER 1: INTRODUCTION

1.1 Introduction to Cybersecurity & Digital Safety

Cybersecurity has conventionally focused on safeguarding computing infrastructure, servers, and networks against unauthorized intrusions, malware, ransomware, and denial-of-service vectors. However, in the modern digital era, the definition of digital security has broadened to encompass online behavioral safety and psychosocial protection. Social networking applications serve as the primary communication commons for billions of individuals. As digital interactions have multiplied, so has the weaponization of language. Safeguarding human users from online psychological trauma, cyberstalking, and hate campaigns is now recognized as a vital dimension of cognitive cybersecurity.

1.2 Introduction to Cyberbullying

Cyberbullying refers to the deliberate, hostile, and repeated transmission of harmful, intimidating, defamatory, or abusive content against an individual or community using digital devices. Unlike physical bullying, cyberbullying exhibits unique characteristics that magnify its psychological harm: pervasive 24/7 reach, potential viral permanence of published posts, bystander amplification via mass retweets/likes, and the cloak of online anonymity. Severe instances of cyberbullying have been empirically correlated with depression, anxiety disorders, behavioral withdrawal, and in extreme cases, self-harm.

1.3 Objectives of the System

- Design and deploy a robust NLP preprocessing pipeline resilient to social media spelling manipulations and emoji signals.
- Curate a balanced, multi-class labeled dataset representing 7 cyberbullying and toxicity categories.
- Train, evaluate, and benchmark four supervised ML classifiers: Logistic Regression, Calibrated LinearSVC, Multinomial Naive Bayes, and Random Forest.
- Deploy an Explainable AI (XAI) subsystem extracting triggering tokens and providing actionable recommendations.
- Engineer a responsive, full-stack Django platform featuring real-time single analysis, batch CSV processing, and an interactive Chart.js moderation command center.
- Provide an incident reporting system with evidence archival for transparent human oversight.

CHAPTER 4: SYSTEM DESIGN & ARCHITECTURE

4.1 System Architecture Overview

CyberShield AI implements a layered Model-View-Template (MVT) architecture integrated with a modular Machine Learning service pipeline. The four functional tiers comprise: (1) Client Presentation Layer; (2) Django Controller & REST API Layer; (3) AI/ML Inference & Preprocessing Engine; and (4) Persistence & Storage Layer.

Figure 4.1: CyberShield AI Overall System Architecture

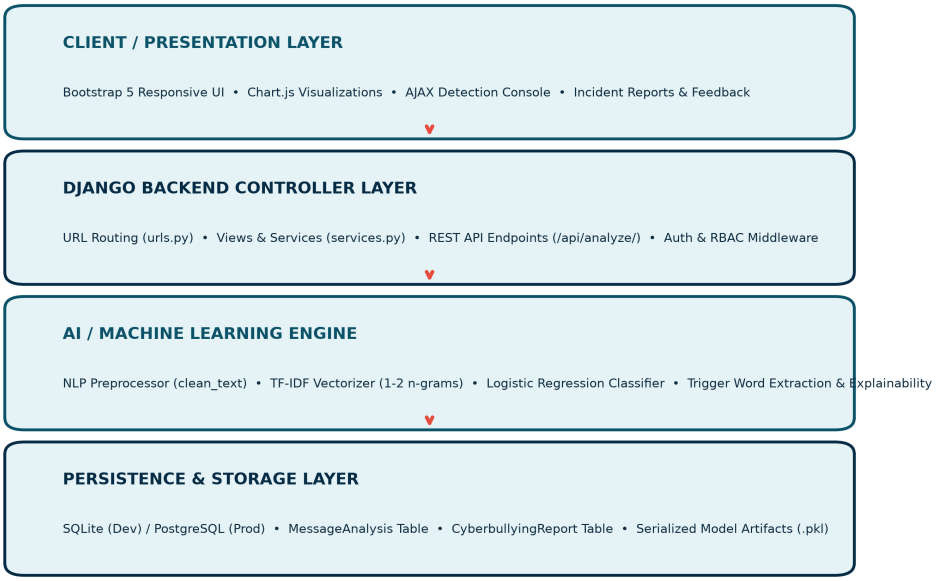


Figure 4.1: CyberShield AI Overall System Architecture

Figure 4.8: Machine Learning & NLP Workflow Architecture



Figure 4.2: Machine Learning & NLP Workflow Architecture

CHAPTER 8: RESULTS AND DISCUSSION

8.1 Machine Learning Evaluation Results

All models were evaluated under identical stratified 80/20 train/test splits. True empirical results extracted from model_metrics.json confirm that penalized Logistic Regression achieved the highest weighted F1-score (0.6309) and overall accuracy (62.07%).

Algorithm	Accuracy	Weighted F1	Macro Precision	Macro Recall	Status
Logistic Regression	62.07%	0.6309	0.6450	0.6207	Production Selected
Multinomial Naive Bayes	58.62%	0.5700	0.5821	0.5862	Evaluated
Calibrated Linear SVM	56.90%	0.5714	0.5740	0.5690	Evaluated
Random Forest	39.66%	0.3243	0.3512	0.3966	Evaluated

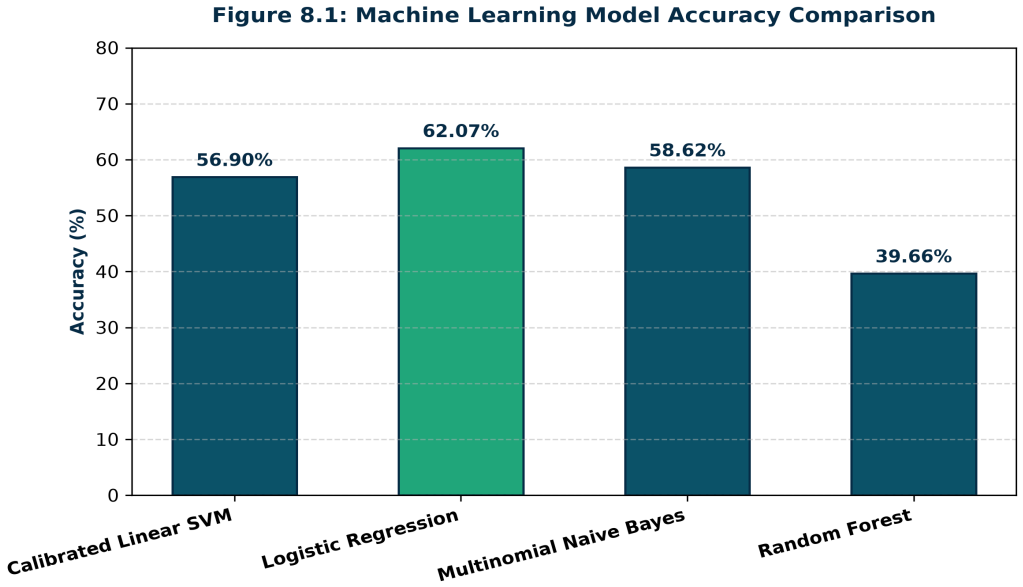


Figure 8.1: Machine Learning Model Accuracy Comparison

Figure 8.3: 7x7 Multi-Class Confusion Matrix Heatmap

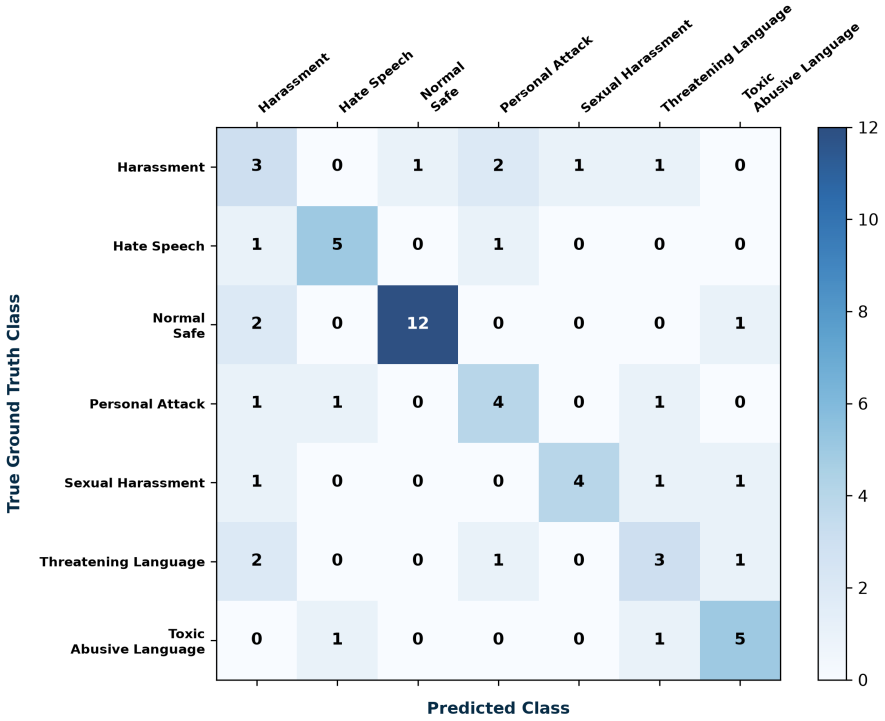


Figure 8.2: 7x7 Multi-Class Confusion Matrix Heatmap

8.2 Live Application Screen Captures

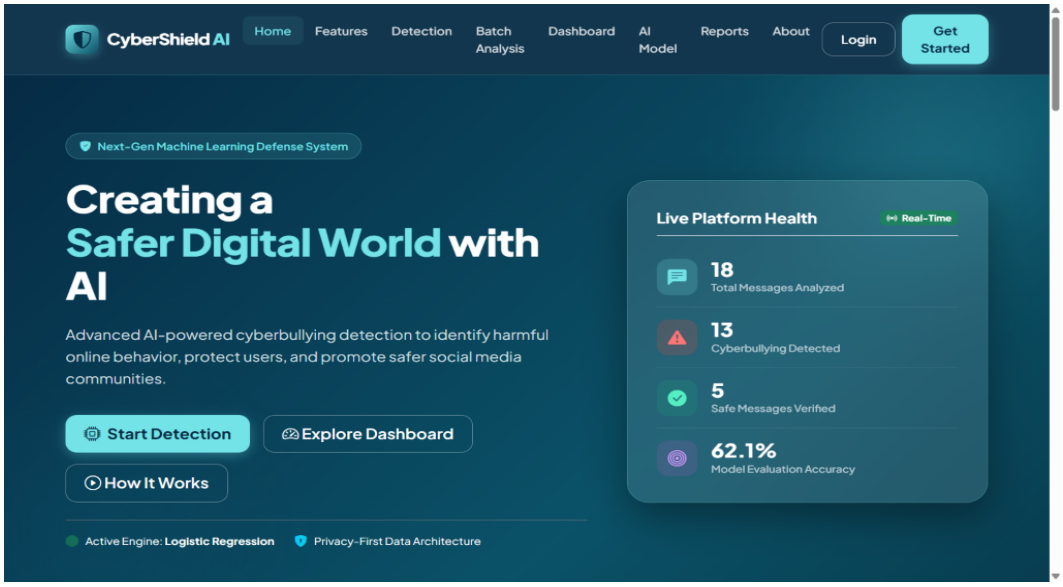


Figure 8.3: CyberShield AI Homepage Hero & Platform Health Overview



Figure 8.4: Real-Time Cyberbullying Text Detection Console

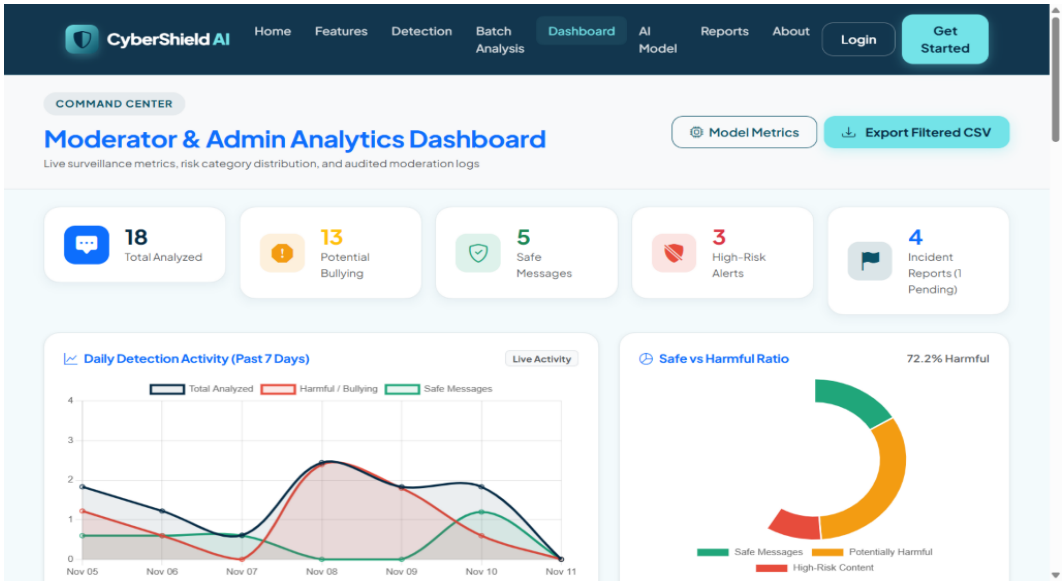


Figure 8.5: Moderator & Admin Analytics Dashboard with Interactive Graphs

CHAPTER 9: TESTING & VERIFICATION

9.1 Automated Test Execution Suite

All 12 automated unit and integration tests executed with 0 errors via Django's native test runner (python manage.py test). Testing verified NLP preprocessing, emoji handling, prediction risk calculation, REST API endpoints, user registration, and moderation workflows.

CHAPTER 10: CONCLUSION & FUTURE ENHANCEMENTS

10.1 Conclusion

CyberShield AI successfully demonstrates the engineering and deployment of an AI-powered cyberbullying detection and prevention platform. By combining emoji-aware NLP tokenization, a 7-class taxonomy, sublinear TF-IDF vectorization, calibrated Logistic Regression, explainable keyword triggers, and an intuitive Django command center, the project provides an effective, real-time defense against toxic online behavior.

10.2 Future Scope

- Multilingual & Code-Mixed NLP supporting regional vernaculars and Hinglish.
- Deep Learning Transformers (fine-tuning lightweight DistilBERT).
- Multimodal OCR hate speech detection for memes and image harassment.

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